HIREMIND · CS4085 DEEP LEARNING

Can a smaller CV–job classifier still recognize suitable candidates?

Efficient CV–job fit classification through distillation and quantization



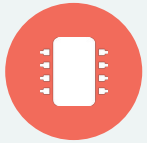
Rawan Khojah (S22107688) · Hams Aljohani (S20106833)
Instructor: Dr. Naila Marir | Effat University · Fall 2026

Screening needs a model that is accurate and cheap



Too many CVs

Recruiters compare every CV with the job by hand, which is slow.



Large models are costly

Big AI models need a GPU or send CV text to an outside service.



Goal: small and local

A small model on a normal CPU is cheaper and keeps CVs private.

Can a small model still recognize suitable candidates?

Two contributions, two fixed hypotheses

Can teacher-guided training improve a compact CV–job fit classifier, and can quantization cut its cost while keeping useful performance?

1 Deep learning: distillation

A large teacher (BERT-base) guides the small student during training.

H1: M1 gains ≥ +1 macro-F1 point over M0.

2 Efficiency: quantization

M1's weights are stored in 8 bits after training.

H2: M2 uses ≤ 40% of M1's storage, loses ≤ 1 point, keeps ≥ half of M1's gain.

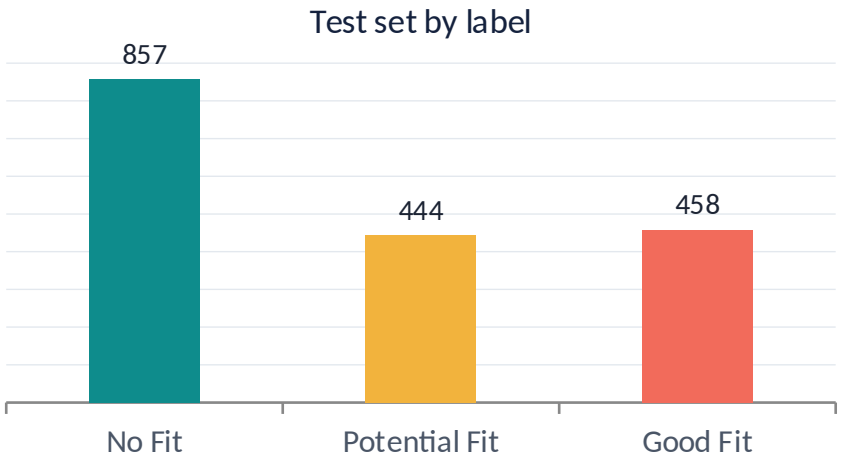
DATA

A job can appear in only one partition

4,993 training pairs

1,162 validation pairs

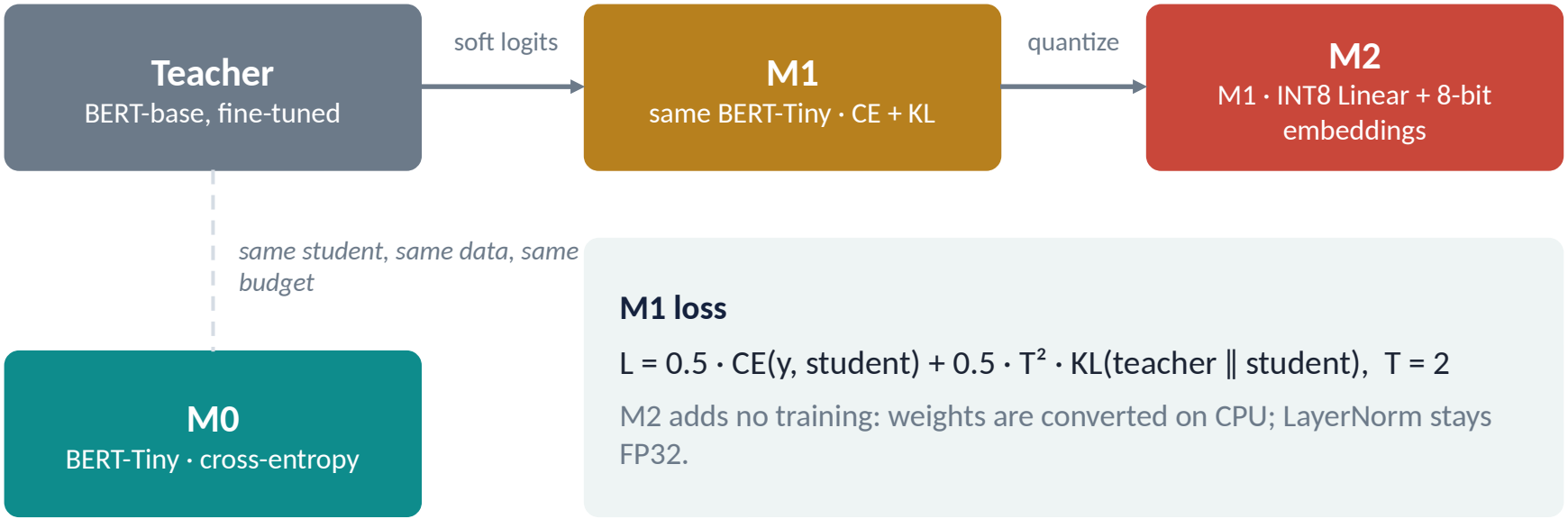
1,759 held-out test pairs



Generalization is to unseen jobs: 474 CVs recur across training and test. Inputs are cut to 256 tokens (median pair 1,563).

METHOD

Three models separate training from compression



M1 loss

$L = 0.5 \cdot CE(y, \text{student}) + 0.5 \cdot T^2 \cdot KL(\text{teacher} \,\|\, \text{student}),\ T = 2$

M2 adds no training: weights are converted on CPU; LayerNorm stays FP32.

EXPERIMENTAL DESIGN

Compare checkpoints under matched conditions



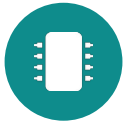
Three paired seeds

42, 43, 44 · same data order and update budget (939 updates, batch 16)



Equal search effort

Four validation-search trials per student method · test untouched until frozen



One CPU benchmark

1 thread · batch 1 · 256 tokens · 30 warm-ups · 200 timed calls · median and IQR

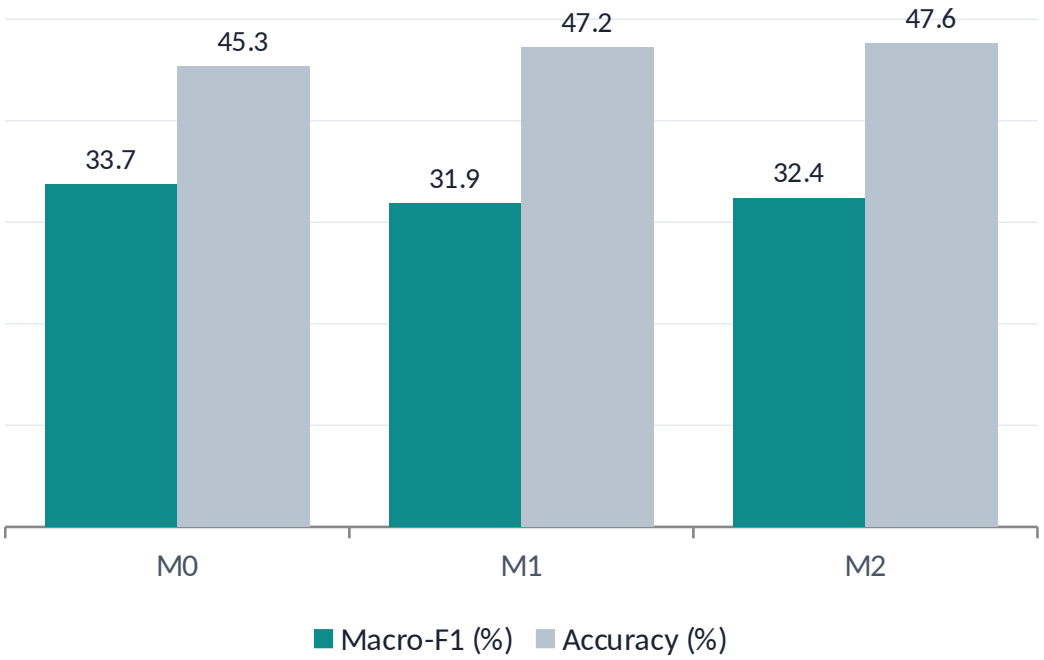


Teacher recovery

Recovered Teacher reproduced all 4,993 cached logits exactly (max difference 0.0)

HELD-OUT TEST

M0 has the highest mean macro-F1



33.70%

M0 mean test macro-F1, the best of the three

Accuracy rises from M0 to M2, but class-balanced macro-F1 falls.

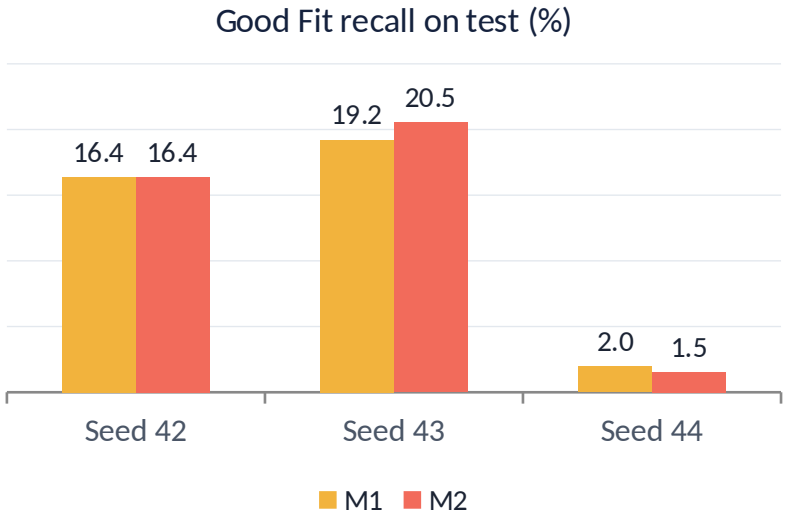
Mean of 3 seeds on 1,759 test pairs; SD 3.9 to 5.2 points.

ERROR ANALYSIS

At seed 44, M2 found only 7 of 458 Good Fit pairs

True ↓ Pred →	No Fit	Potential	Good Fit
No Fit	798	13	46
Potential	356	39	49
Good Fit	419	32	**7**

M2, seed 44, held-out test (rows are true labels)



419 Good Fit pairs were called No Fit. M1 already failed at seed 44, so quantization did not cause it.

EFFICIENCY

Smaller model, but slower on this CPU

−73.4%

saved model size

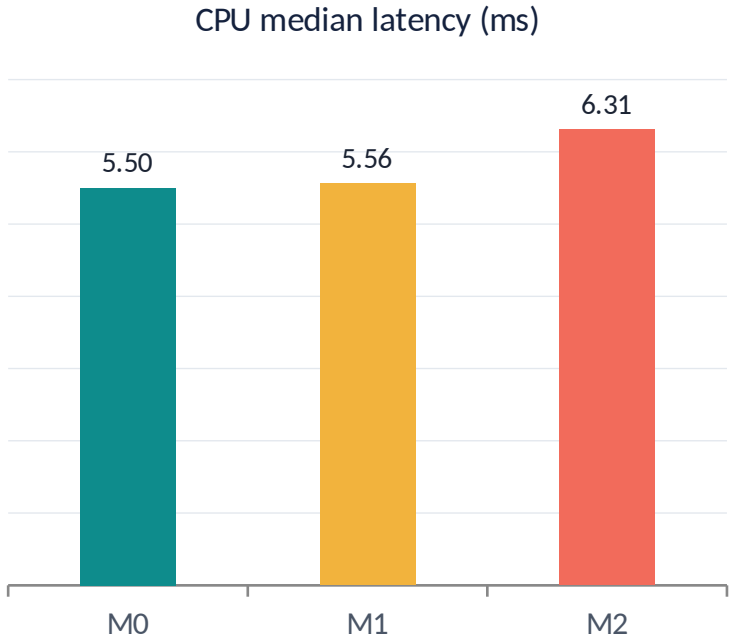
17.56 → 4.67 MB (M1 → M2)

+13.5%

CPU latency

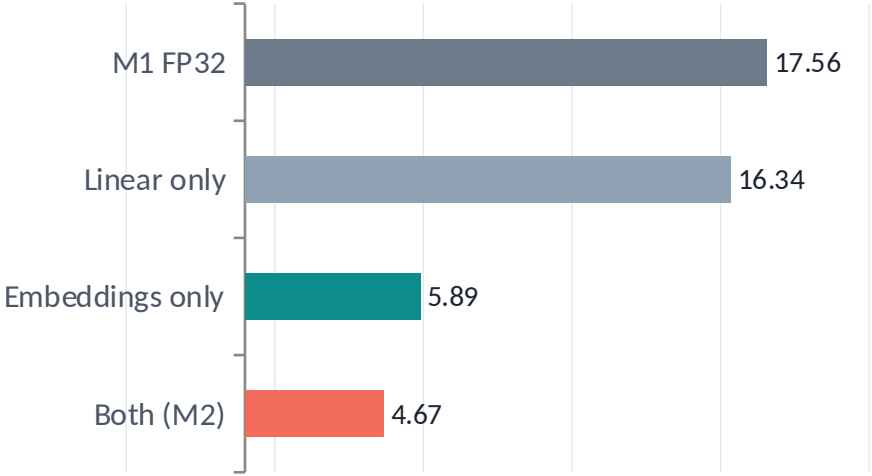
5.56 → 6.31 ms per prediction

Parameters (4.39 M) and MACs (0.134 G) do not change: quantization changes how weights are stored, not how many there are.



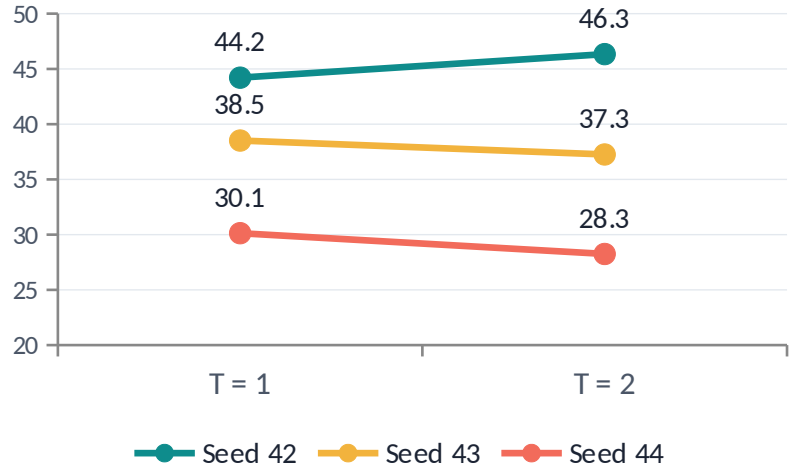
Embeddings account for most of the storage saving

Quantization variant: saved size (MB)



Validation macro-F1 barely moves: 37.28 → 37.20 → 37.10 → 36.96

Distillation temperature: validation macro-F1



Temperature changes direction across seeds: no reliable advantage.

HYPOTHESES

H1 fails; two H2 criteria are met

H1 · distillation gain ≥ +1	M1 − M0 = −1.82 macro-F1 points	NOT MET
H2 · storage ≤ 40% of M1	Size ratio 0.266 for every seed	MET
H2 · quality loss ≤ 1 point	M2 − M1 = +0.51 macro-F1 points	MET
H2 · keeps ≥ half of M1's gain	No gain to keep, because H1 failed	NOT TESTABLE

LIMITATIONS

What our results cannot show

Three seeds, one teacher
Small differences between models are not reliable.

Familiar CVs
474 CVs are in training and test: results hold for new jobs, not new candidates.

Only 256 tokens
The model sees the start of each pair; the median pair has 1,563 tokens.

One CPU
Speed results may differ on other hardware.

Unknown label source
The dataset does not say how its labels were made.

Not tested for hiring
No fairness, calibration or real deployment test yet.

SENIOR 2

Keep recruiter review in the decision loop



Advisory signal only

The classifier runs locally on CPU; it never rejects a candidate.

Weak Good Fit recall

A Good Fit miss can be routed to HR review instead of being dropped.

Still to test

Deployment, fairness and calibration need separate evaluation.

CONCLUSION

Storage improved. Recognizing suitable candidates remains harder.

1 **Baseline wins**
M0 has the best mean test macro-F1 (33.70%); distillation did not help.

2 **Smaller, not faster**
Quantization cut saved size by 73% with no quality loss, but ran 13% slower.

3 **Next study**
Longer inputs (512 tokens) and better Good Fit recall on a new evaluation cohort.

Thank you · Questions?

github.com/Hams770/HireMind-DL